**EXP NO:13 Experiment to handle missing values and inappropriate data in a given data set.**

**AIM:**
**To understand the procedure to handle missing and inappropriate data.**

**PROGRAM:**

**python**

```
import numpy as np
import pandas as pd

# Read dataset
df = pd.read_csv("/content/Movie_Booking_Dataset.csv")
df

# Identify duplicate rows
df.duplicated()

# Display information
df.info()

# Remove duplicate rows
df.drop_duplicates(inplace=True)
df

# Check number of rows
len(df)

# Reset index
df.reset_index(drop=True, inplace=True)
df

# Remove duplicate Age Group column
df.drop(columns=['Age_Group.1'], inplace=True, errors='ignore')
df

# Check unique age groups
df.Age_Group.unique()

# Check unique cinemas
df.Cinema.unique()

# Correct inappropriate cinema name
df['Cinema'] = df['Cinema'].replace('PVRR', 'PVR')

# Replace inappropriate negative values with NaN
df.loc[df['EstimatedIncome'] < 0, 'EstimatedIncome'] = np.nan
df.loc[df['TicketBill'] < 0, 'TicketBill'] = np.nan

# Correct inappropriate NoOfPax values
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan

# Correct inappropriate ratings
df.loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5), 'Rating(1-5)'] = np.nan

# Check food preferences
df.SnackPreference.unique()

# Standardize food preference
df['SnackPreference'] = df['SnackPreference'].replace(
    {'Vegetarian': 'Veg', 'veg': 'Veg', 'non-Veg': 'Non-Veg'}
)

df.SnackPreference.unique()

# Fill continuous numerical missing values using mean
df['EstimatedIncome'] =
df['EstimatedIncome'].fillna(round(df['EstimatedIncome'].mean()))
df['TicketBill'] = df['TicketBill'].fillna(round(df['TicketBill'].mean()))

# Fill discrete numerical missing values using median
df['NoOfPax'] = df['NoOfPax'].fillna(round(df['NoOfPax'].median()))
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()))

# Final cleaned dataset
df
```

**OUTPUT:**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   BookingID        11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Rating(1-5)      11 non-null     int64
 3   Cinema           11 non-null     object
 4   SnackPreference  11 non-null     object
 5   TicketBill       10 non-null     float64
 6   NoOfPax          11 non-null     int64
 7   EstimatedIncome  11 non-null     int64
 8   Age_Group.1      11 non-null     object
dtypes: float64(1), int64(4), object(4)
memory usage: 924.0+ bytes
```

| | BookingID | Age_Group | Rating(1-5) | Cinema | SnackPreference | TicketBill | NoOfPax | EstimatedIncome | Age_Group.1 |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 18-25 | 4 | PVR | Veg | 450.0 | 2 | 35000 | 18-25 |
| **1** | 2 | 26-30 | 5 | INOX | Non-Veg | 720.0 | 3 | 52000 | 26-30 |
| **2** | 3 | 31-35 | 6 | Cinepolis | Veg | 600.0 | 2 | 48000 | 31-35 |
| **3** | 4 | 18-25 | -1 | INOX | Vegetarian | 500.0 | 2 | 65000 | 18-25 |
| **4** | 5 | 35+ | 3 | PVR | Vegetarian | 800.0 | 2 | 45000 | 35+ |
| **5** | 6 | 35+ | 4 | PVRR | Non-Veg | 950.0 | 2 | 72000 | 35+ |
| **6** | 7 | 31-35 | 4 | Cinepolis | Veg | 1000.0 | -1 | 58000 | 31-35 |
| **7** | 8 | 18-25 | 7 | INOX | veg | 1200.0 | -10 | 90000 | 18-25 |
| **8** | 9 | 26-30 | 2 | PVR | Non-Veg | 1100.0 | 3 | -9999 | 26-30 |
| **9** | 10 | 31-35 | 5 | Cinepolis | non-Veg | NaN | 4 | 67000 | 31-35 |

**RESULT:**

**Thus the program to handle missing values and inappropriate data in the given data set was executed successfully.**